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(19) **United States**(12) **Patent Application Publication**
KIM et al.(10) **Pub. No.: US 2022/0165650 A1**(43) **Pub. Date: May 26, 2022**(54) **PACKAGING SUBSTRATE AND
SEMICONDUCTOR APPARATUS
COMPRISING SAME**(71) Applicant: **ABSOLICS INC.**, Covington, GA (US)(72) Inventors: **Sungjin KIM**, SUWANEE, GA (US);
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Byungkyu JANG, Suwon-si (KR)(73) Assignee: **ABSOLICS INC.**, Covington, GA (US)(21) Appl. No.: **17/433,347**(22) PCT Filed: **Mar. 6, 2020**(86) PCT No.: **PCT/KR2020/003175**

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7, 2019, provisional application No. 62/816,965, filed
on Mar. 12, 2019.**Publication Classification**(51) **Int. Cl.****H01L 23/498** (2006.01)**H01L 23/15** (2006.01)(52) **U.S. Cl.****CPC H01L 23/49822** (2013.01); **H01L 25/0652**
(2013.01); **H01L 23/49838** (2013.01); **H01L**
23/15 (2013.01)

(57)

ABSTRACT

The embodiment relates to a packaging substrate and a semiconductor apparatus, including an element unit including a semiconductor element; and a packaging substrate electrically connected to the element unit; and it applies a glass substrate as a core of the packaging substrate, thereby can significantly improve electrical properties such as a signal transmission rate by connecting the semiconductor element and a motherboard to be closer to each other so that electrical signals are transmitted through as short a path as possible. Therefore, it can significantly improve electrical properties such a signal transmission rate, substantially prevent generating of parasitic element, and simplify a process of treatment for an insulating layer, and thus provides a packaging substrate applicable to a high-speed circuit.

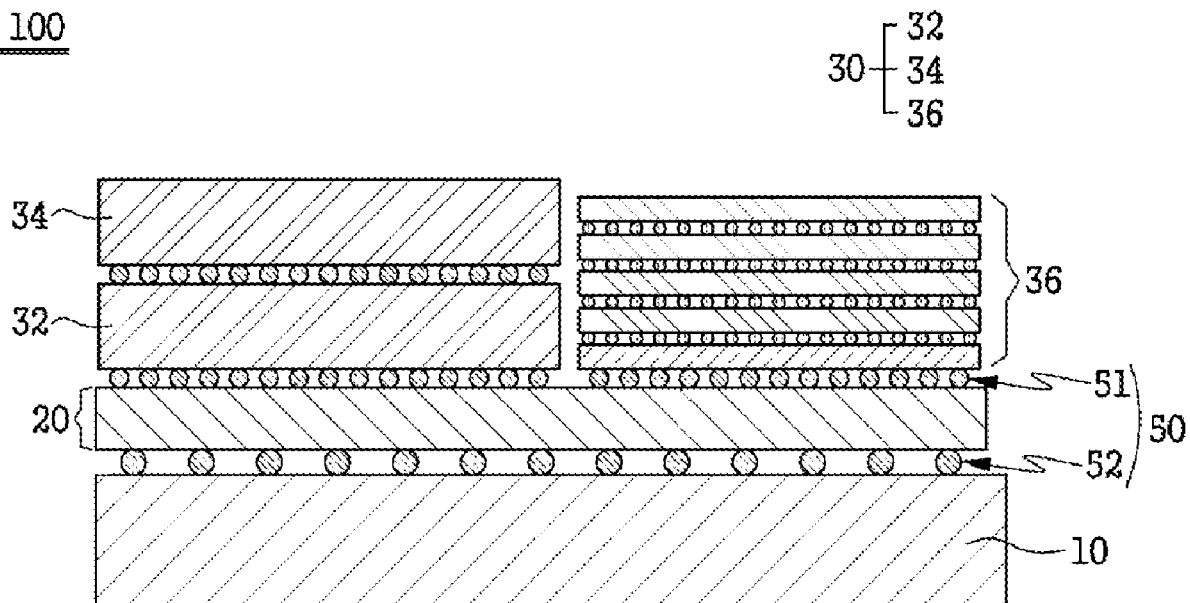
100

FIG. 1

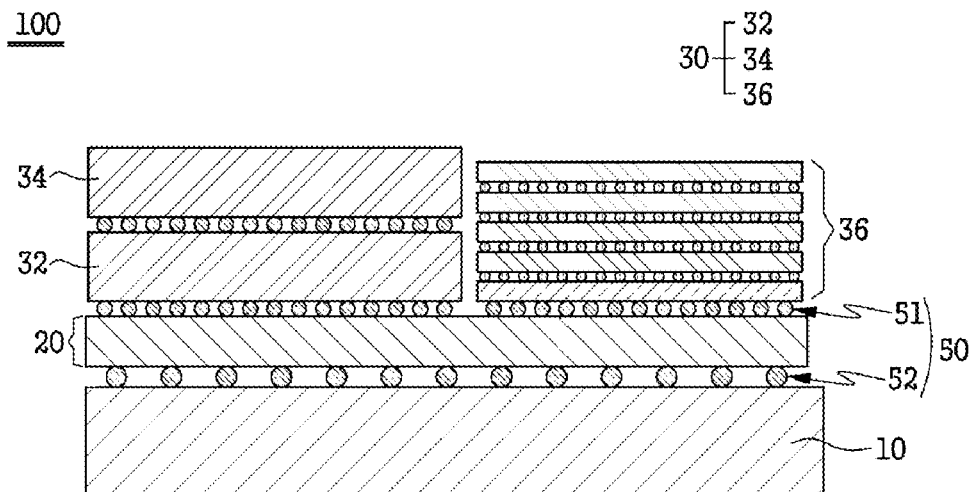


FIG. 2

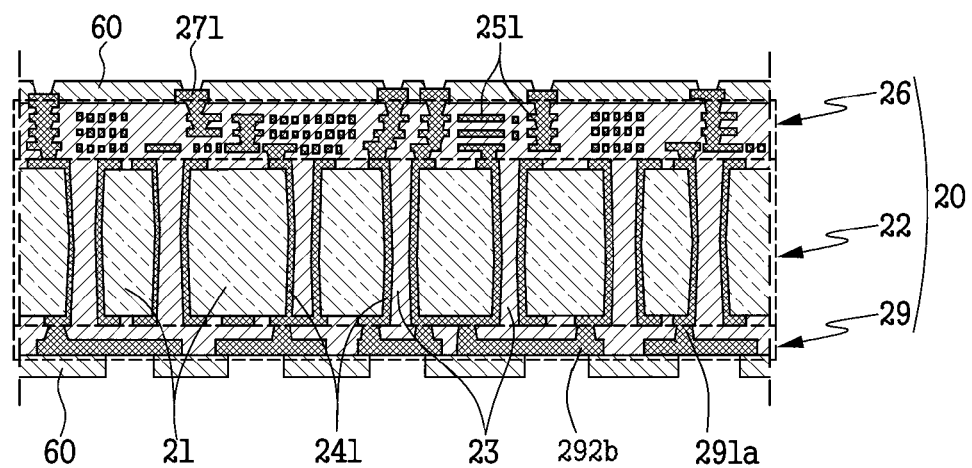


FIG. 3A

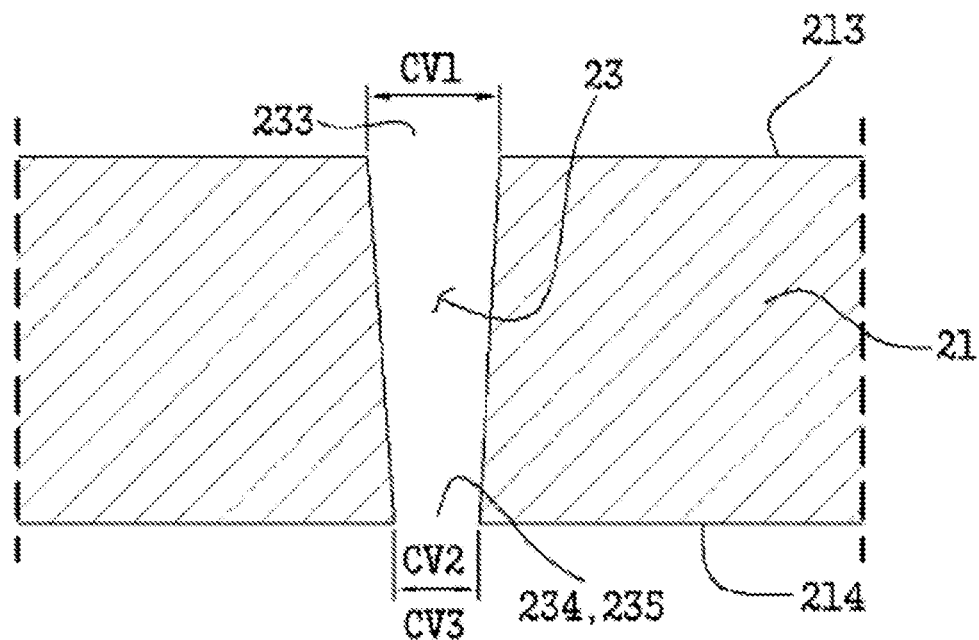


FIG. 3B

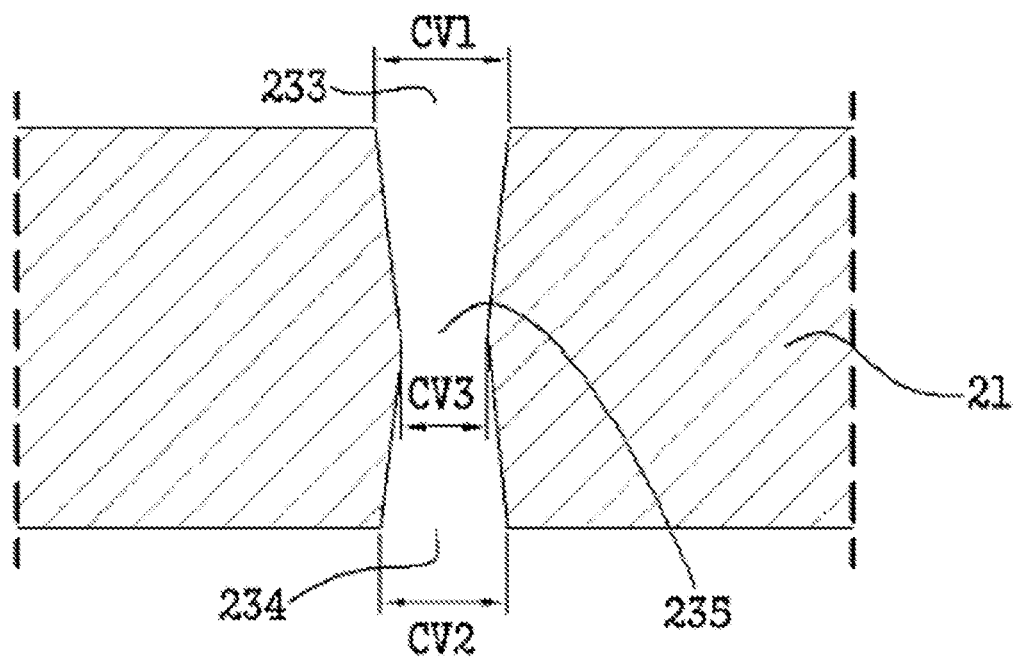


FIG. 4

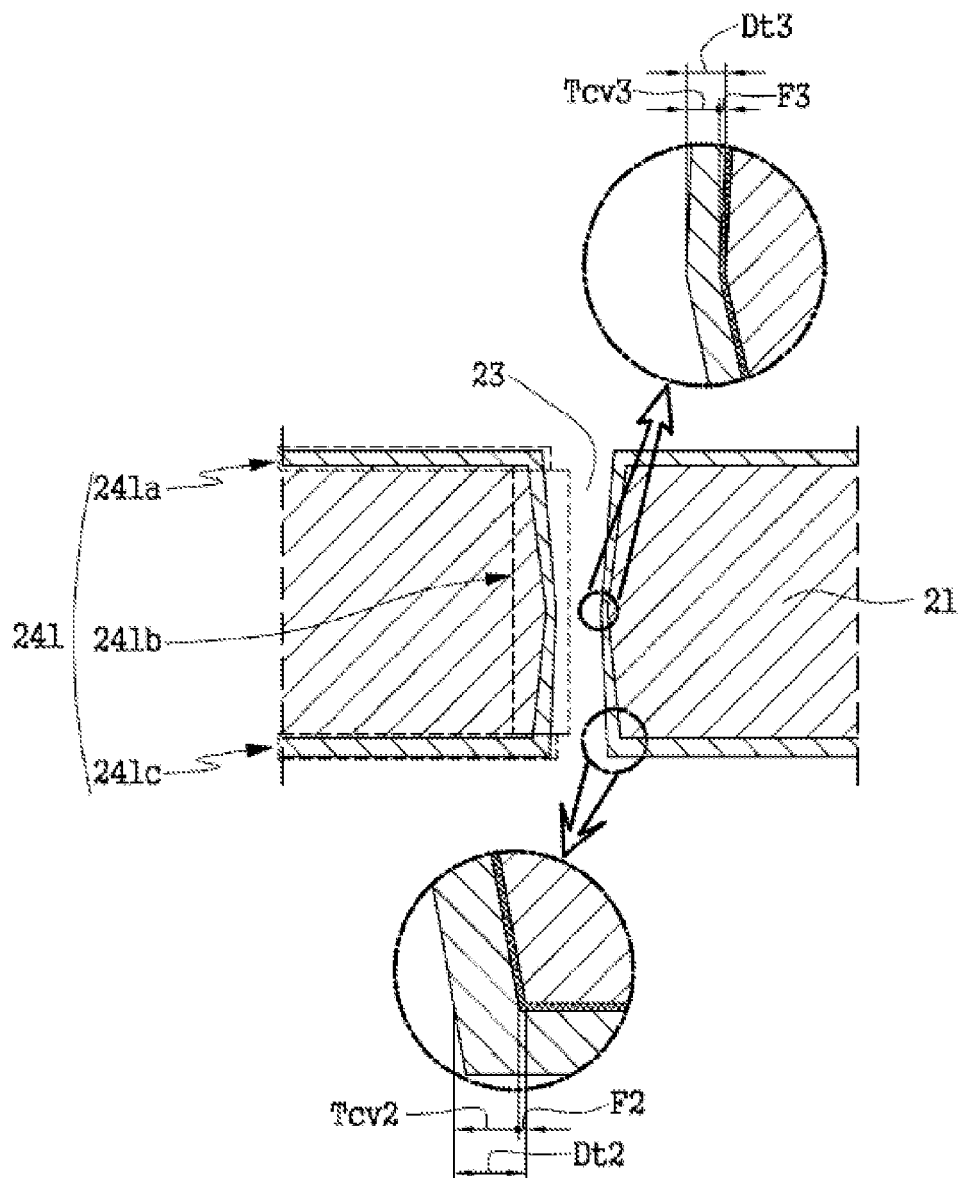


FIG. 5

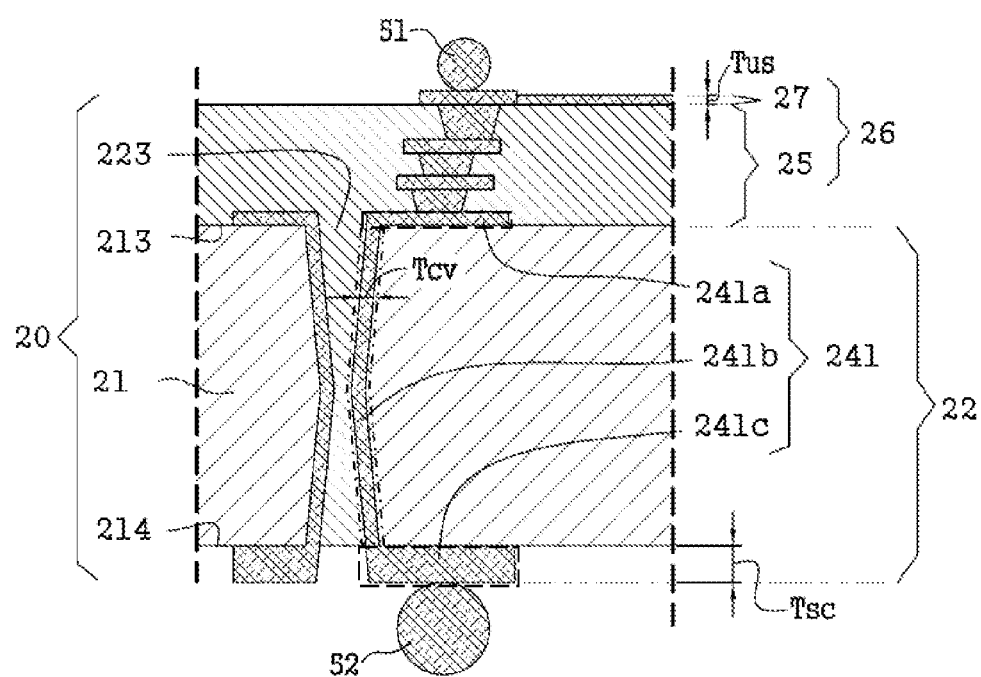


FIG. 6

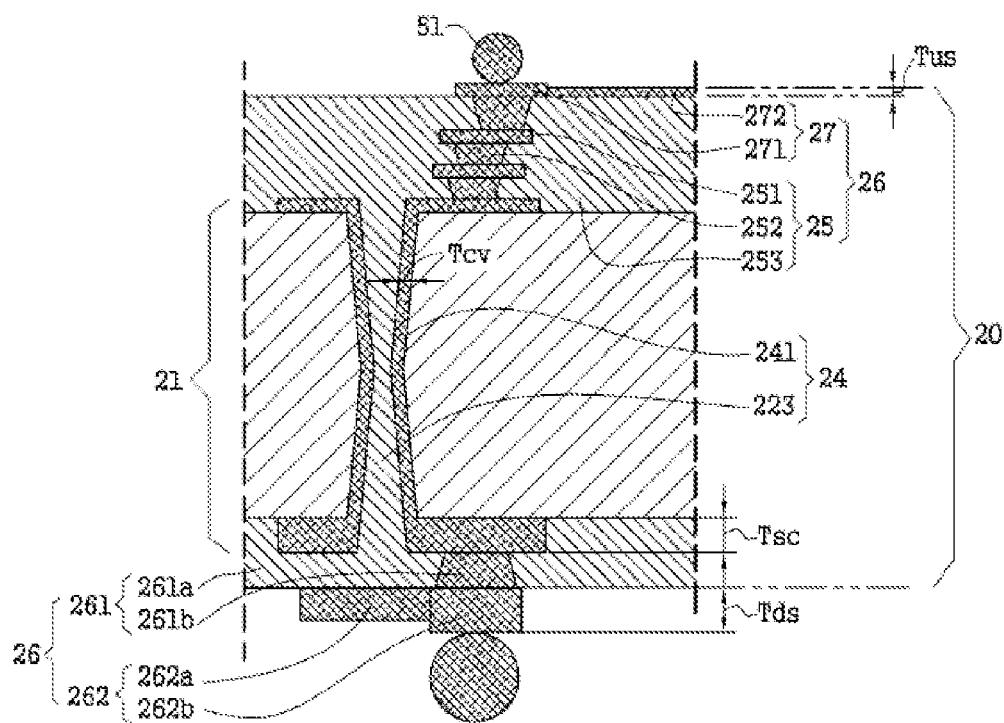


FIG. 7A

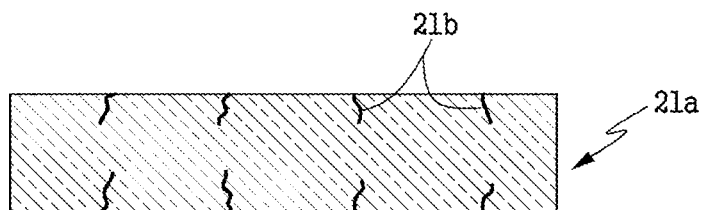


FIG. 7B

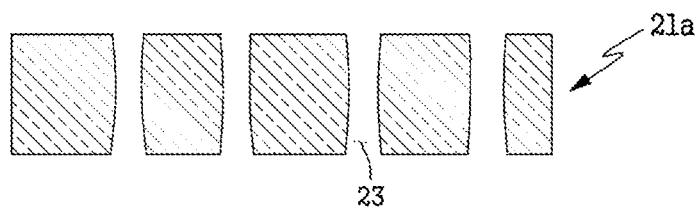


FIG. 7C

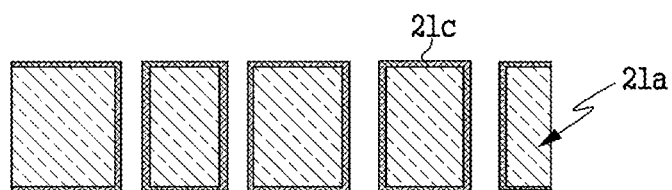


FIG. 7D

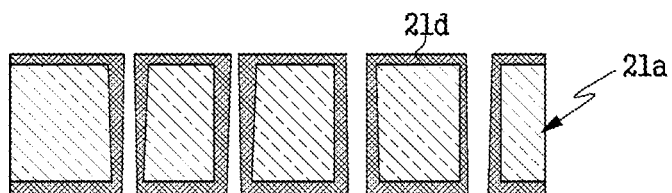


FIG. 7E

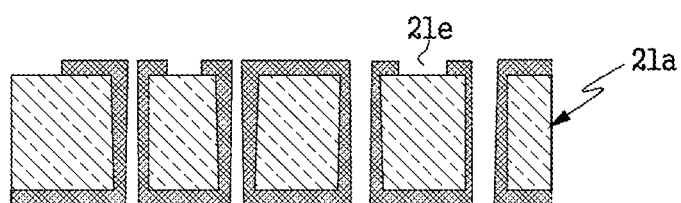


FIG. 8A

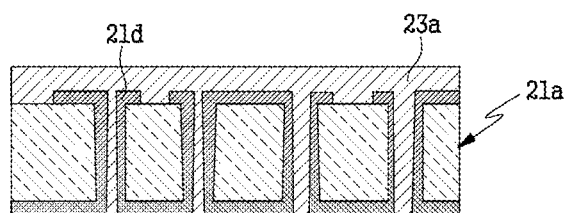


FIG. 8B

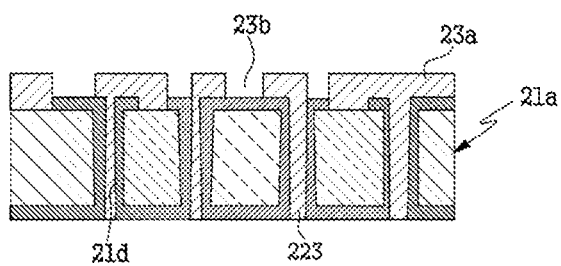


FIG. 8C

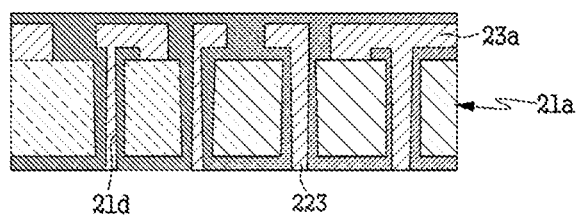


FIG. 8D

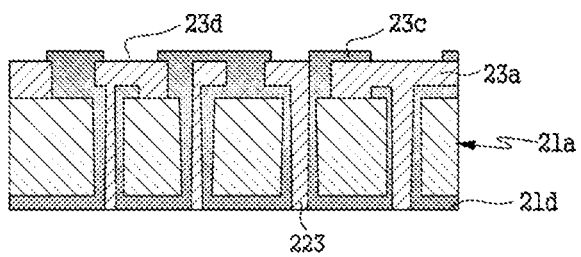


FIG. 8E

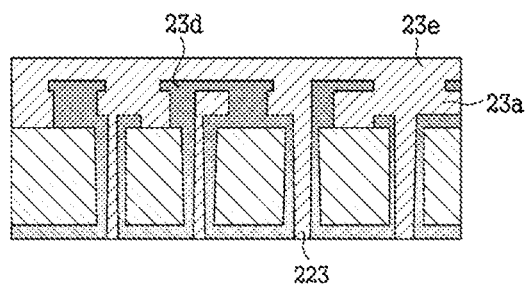


FIG. 9A

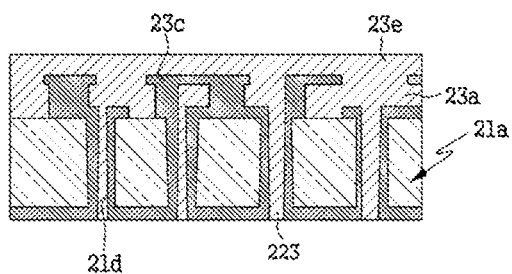


FIG. 9B

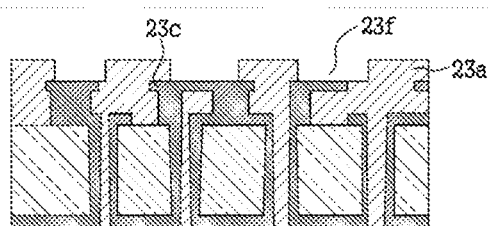


FIG. 9C

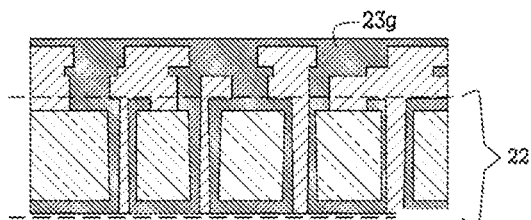


FIG. 9D

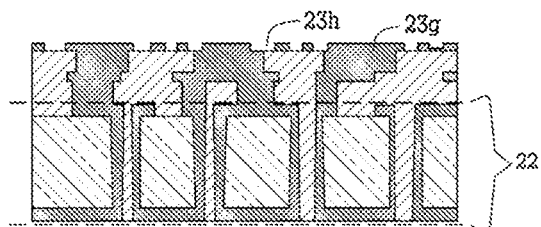
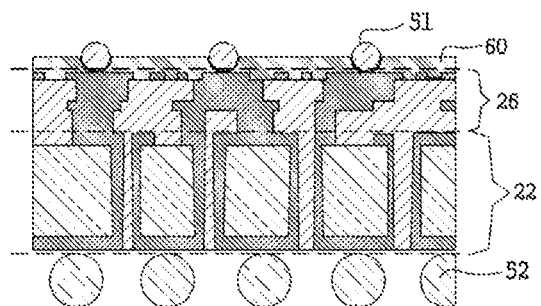


FIG. 9E



PACKAGING SUBSTRATE AND SEMICONDUCTOR APPARATUS COMPRISING SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to U.S. Provisional Patent Application No. 62/814,941, filed on Mar. 7, 2019, and US Provisional Patent Application No. 62/816,965, filed on Mar. 12, 2019, and all the benefits accruing therefrom under the priority, the content of which in their entireties are herein incorporated by reference.

FIELD

[0002] The embodiments relate to packaging substrate, and semiconductor apparatus comprising same.

RELATED ART

[0003] In the manufacturing of electronic components, the implementation of a circuit on a semiconductor wafer is referred to as a Front-End Process (FE), and the assembly of a wafer such that it can be actually used in a product is referred to as a Back-End Process (BE). A packaging process is included in the Back-End process.

[0004] Four key technologies of the semiconductor industry that enable the rapid development of electronic products in recent years include semiconductor technology, semiconductor packaging technology, manufacturing process technology, and software technology. Semiconductor technology has been developed in various forms such as line width of a nanometer unit, which is smaller than a micrometer unit, 10 million or more cells, high-speed operation, and much heat dissipation, but technology of packaging it completely is not supported yet. Thus, the electrical performance of semiconductors may be determined by the packaging technology and the resulting electrical connection rather than the performance of the semiconductor itself.

[0005] Ceramic or resin is used as the material of a packaging substrate. In the case of a ceramic substrate such as Si substrate, it is not easy to mount a high-performance and high-frequency semiconductor element thereon due to a high resistance or high dielectric constant. In the case of a resin substrate, it is possible to mount a high-performance and high-frequency semiconductor element thereon, but there is a distinct limitation to the reduction of pitches of wirings.

[0006] Recently, research is being conducted to apply silicon or glass to a high-end packaging substrate. By forming a through-via on a silicon or glass substrate and applying a conductive material into the through-via, it is possible to shorten a length of conductive lines between an element and a motherboard, and have excellent electric characteristics.

[0007] As related art documents, there are

[0008] Korean Patent Publication No. 10-2019-0008103,

[0009] Korean Patent Publication No. 10-2016-0114710,

[0010] Korean Patent No. 10-1468680, and the like.

DISCLOSURE

Technical Problem

[0011] The objective of the embodiment is to provide a more integrated packaging substrate and a semiconductor apparatus comprising same, by applying a glass substrate.

Technical Solution

[0012] To achieve the above objective, a semiconductor apparatus according to one embodiment includes:

[0013] a semiconductor element unit where one or more semiconductor elements are disposed; a packaging substrate electrically connected to the semiconductor element; and a motherboard electrically connected to the packaging substrate, transmitting electrical signals of the semiconductor element and external, and connecting each other;

[0014] wherein a packaging substrate includes a core layer, and an upper layer disposed on the core layer,

[0015] the core layer includes a glass substrate and a core via,

[0016] the glass substrate is with a first surface and a second surface facing each other,

[0017] the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

[0018] the core layer includes a core distribution layer disposed on a surface of the glass substrate or the core via,

[0019] the core distribution layer includes an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, and

[0020] the upper layer is disposed on the first surface and includes an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

[0021] wherein the core via has a first surface opening part diameter which is a diameter at an opening part in contact with the first surface, a second surface opening part diameter which is a diameter at an opening part in contact with the second surface, and a minimum inner diameter which is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

[0022] wherein at a position of the minimum inner diameter, a thickness of the electrically conductive layer is 90% or more, when a distance from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via is 100% as total.

[0023] In one embodiment, at an opening part where a larger one between the first surface opening part diameter and the second surface opening part diameter is disposed, a thickness of the electrically conductive layer may be 90% or more, when a distance from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total.

[0024] In one embodiment, an upper insulating layer and an upper distribution pattern may be included,

[0025] wherein the upper insulating layer may be disposed on the first surface,

[0026] the upper distribution pattern may be an electrically conductive layer at least a part of which is electrically connected to the core distribution layer, and built in the upper insulating layer,

[0027] the upper distribution pattern may be at least partially including a fine pattern, and

[0028] the fine pattern may have a width and an interval of less than 4 μm , respectively.

[0029] In one embodiment, the packaging substrate may have a resistance value of about $27.5 \times 10^{-6} \Omega$ or less, based on a cut one of the packaging substrate into 100 $\mu\text{m} \times 100 \mu\text{m}$ as an upper surface size.

[0030] To achieve the above objective, a packaging substrate according to one embodiment includes:

[0031] a core layer, and an upper layer disposed on the core layer,

[0032] wherein the core layer includes a glass substrate and a core via,

[0033] the glass substrate is with a first surface and a second surface facing each other,

[0034] the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

[0035] the core layer includes a core distribution layer disposed on a surface of the glass substrate or the core via,

[0036] the core distribution layer includes an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and the second surface through the core via, and

[0037] the upper layer is disposed on the first surface and includes an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

[0038] wherein the core via has a first surface opening part diameter which is a diameter at an opening part in contact with the first surface, a second surface opening part diameter which is a diameter at an opening part in contact with the second surface, and a minimum inner diameter which is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

[0039] wherein a thickness of a thinner one among electrically conductive layers of the core distribution layer, may be the same as or thicker than a width of a thinner one among electrically conductive layers of the upper layer.

[0040] To achieve the above objective, a semiconductor element according to another embodiment includes:

[0041] a semiconductor element unit where one or more semiconductor elements are disposed; a packaging substrate electrically connected to the semiconductor element; and a motherboard electrically connected to the packaging substrate, transmitting electrical signals of the semiconductor element and external, and connecting each other;

[0042] wherein a packaging substrate includes a core layer, and an upper layer disposed on the core layer,

[0043] the core layer includes a glass substrate and a core via,

[0044] the glass substrate is with a first surface and a second surface facing each other,

[0045] the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

[0046] the core layer includes a core distribution layer disposed on a surface of the glass substrate or the core via,

[0047] the core distribution layer includes an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and the second surface through the core via, and

[0048] the upper layer is disposed on the first surface and includes an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

[0049] wherein the core distribution layer includes a first surface core pattern which is an electrically conductive layer disposed on at least a part of the first surface, a second surface core pattern which is an electrically conductive layer disposed on at least a part of the second surface, and a core via pattern which is an electrically conductive layer electrically connecting the first surface core pattern and the second surface core pattern through the core via,

[0050] wherein the core via has a first surface opening part diameter which is a diameter at an opening part in contact with the first surface, a second surface opening part diameter which is a diameter at an opening part in contact with the second surface, and a minimum inner diameter which is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

[0051] wherein the core via pattern may have an average distance of 1 μm or less, between one surface of the core via pattern close to an inner diameter surface of the core via; and the inner diameter surface of the core via.

[0052] In one embodiment, at a position of the minimum inner diameter, a thickness of the electrically conductive layer may be 90% or more, when a distance from an inner diameter surface of the core via to a surface of the electrically conductive layer of the core via is 100% as total.

[0053] In one embodiment, the packaging substrate may have a resistance value of about $27.5 \times 10^{-6} \Omega$ or less, based on an upper surface of the one being cut into the size of 100 $\mu\text{m} \times 100 \mu\text{m}$.

[0054] In one embodiment, an upper insulating layer and an upper distribution pattern may be included,

[0055] wherein the upper insulating layer may be disposed on the first surface,

[0056] the upper distribution pattern may be an electrically conductive layer at least a part of which is electrically connected to the core distribution layer, and built in the upper insulating layer,

[0057] the upper distribution layer may be at least partially including a fine pattern, and

[0058] the fine pattern may have a width and an interval of less than 4 μm , respectively.

[0059] To achieve the above objective, a packaging substrate according to another embodiment includes:

[0060] a core layer, and an upper layer disposed on the core layer,

[0061] wherein the core layer includes a glass substrate and a core via,

[0062] the glass substrate is with a first surface and a second surface facing each other,

[0063] the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

[0064] the core layer includes a core distribution layer disposed on a surface of the glass substrate or the core via,

[0065] the core distribution layer includes an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and the second surface through the core via, and

[0066] the upper layer is disposed on the first surface and includes an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

[0067] wherein the core distribution layer includes a first surface core pattern which is an electrically conductive layer disposed on at least a part of the first surface, a second surface core pattern which is an electrically conductive layer disposed on at least a part of the second surface, and a core via pattern which is an electrically conductive layer electrically connecting the first surface core pattern and the second surface core pattern through the core via,

[0068] wherein the core via has a first surface opening part diameter which is a diameter at an opening part in contact with the first surface, a second surface opening part diameter which is a diameter at an opening part in contact with the second surface, and a minimum inner diameter which is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

[0069] wherein the core via pattern may have an average distance of 1 μm or less, between one surface of the core via pattern close to an inner diameter surface of the core via; and the inner diameter surface of the core via.

Effects

[0070] Packaging substrate and semiconductor apparatus comprising same of the embodiment can significantly improve electrical properties such as a signal transmission rate by connecting the semiconductor element and a motherboard to be closer to each other so that electrical signals are transmitted through as short a path as possible.

[0071] Also, since a glass substrate applied as a core of substrate is an insulator itself, there is a lower possibility of generating parasitic element compared to a conventional silicon core, and thus it is possible to simplify a process of treatment for an insulating layer and it is also applicable to a high-speed circuit.

[0072] In addition, unlike silicon being manufactured in the form of a round wafer shape, the glass substrate is manufactured in the form of a large panel, and thus mass production is relatively easy and economic efficiency can be further improved.

BRIEF DESCRIPTION OF THE DRAWINGS

[0073] FIG. 1 is a conceptual view for illustrating a cross section of a semiconductor apparatus according to one embodiment.

[0074] FIG. 2 is a conceptual view for illustrating a cross section of a packaging substrate according to another embodiment.

[0075] FIGS. 3 (a) and (b) are conceptual views for illustrating a cross section of a core via applied to the embodiment, respectively.

[0076] FIG. 4 is a conceptual view for illustrating a shape of a core distribution pattern formed on a glass substrate and a thickness of an electrically conductive layer by a cross sections thereof.

[0077] FIG. 5 and FIG. 6 are detailed conceptual views for illustrating a part of cross sections of a packaging substrate according to the embodiment, respectively.

[0078] FIGS. 7 to 9 are flowcharts for illustrating a process of manufacturing a packaging substrate according to the embodiment by using cross sections thereof.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0079] Hereinafter, examples will be described in detail with reference to the accompanying drawings so that they can be easily practiced by those skilled in the art to which the embodiment pertains. However, the embodiment may be embodied in many different forms and is not to be construed as being limited to the embodiments set forth herein. Like reference numerals designate like elements throughout the specification.

[0080] Throughout the present specification, the phrase "combination(s) thereof" included in a Markush-type expression denotes one or more mixtures or combinations selected from the group consisting of components stated in the Markush-type expression, that is, denotes that one or more components selected from the group consisting of the components are included.

[0081] Throughout the present specification, terms such as "first," "second," "A," or "B" are used to distinguish the same terms from each other. The singular forms "a," "an," and "the" include the plural form unless the context clearly dictates otherwise.

[0082] Throughout the present specification, the term "X-based" may mean that a compound includes a compound corresponding to X, or a derivative of X.

[0083] Throughout the present specification, "B being disposed on A" means that B is disposed in direct contact with A or disposed over A with another layer or structure interposed therebetween and thus should not be interpreted as being limited to B being disposed in direct contact with A.

[0084] Throughout the present specification, "B being connected to A" means that B is connected to A directly or through another element therebetween, and thus should not be interpreted as being limited to B being directly connected to A, unless otherwise noted.

[0085] Throughout the present specification, a singular form is contextually interpreted as including a plural form as well as a singular form unless specially stated otherwise.

[0086] The inventors have recognized that, in the process of developing a semiconductor apparatus capable of exhibiting high performance with a more integrated and thinner thickness, not only the apparatus itself but also the packaging process is an important factor for improving its performance. And while researching this, inventors have confirmed that, by applying a glass core in a single layer and controlling the shape of a through-via, an electrically conductive layer formed thereon, etc., it is possible to make a packaging substrate thinner and to improve the electrical properties of the semiconductor apparatus, unlike a conventional interposer and organic substrate in which two or more layers of cores are applied on a motherboard as a packaging substrate, and thereby completed the invention.

[0087] FIG. 1 is a conceptual view for illustrating a cross section of a semiconductor apparatus according to one embodiment, FIG. 2 is a conceptual view for illustrating a cross section of a packaging substrate according to another embodiment, FIGS. 3 (a) and (b) are conceptual views for illustrating a cross section of a core via applied to the embodiment, respectively, FIG. 4 is a conceptual view for

illustrating a shape of a core distribution pattern formed on a glass substrate and a thickness of an electrically conductive layer by a cross sections thereof, and FIG. 5 and FIG. 6 are detailed conceptual views for illustrating a part of cross sections of a packaging substrate according to the embodiment, respectively. Hereinafter, the embodiment will be described in more detail with reference to FIGS. 1 to 5.

[0088] To achieve the above objective, a semiconductor apparatus 100 according to the embodiment includes a semiconductor element unit 30 where one or more semiconductor elements 32, 34, and 36 are disposed; a packaging substrate 20 electrically connected to the semiconductor element; and a motherboard 10 electrically connected to the packaging substrate, transmitting electrical signals of the semiconductor element and external, and connecting each other.

[0089] The packaging substrate 20 according to another embodiment includes a core layer 22 and an upper layer 26.

[0090] The semiconductor element unit 30 refers to the elements mounted on a semiconductor apparatus and is mounted on the packaging substrate 20 through a connecting electrode or the like. In detail, for example, a computation element (a first element 32 and a second element 34) such as a central processing unit (CPU) and a graphics processing unit (GPU), a memory element (a third element 36) such as a memory chip, or the like may be applied as the semiconductor element unit 30, but any semiconductor element capable of being mounted on a semiconductor apparatus may be applicable without limitation.

[0091] A motherboard such as a printed circuit board and a printed wiring board may be applied as the motherboard 10.

[0092] The packaging substrate 20 includes a core layer 22 and an upper layer 26 disposed on one surface of the core layer.

[0093] The packaging substrate 20 may further include a lower layer 29 disposed under the core layer, optionally.

[0094] The core layer 22 includes a glass substrate 21; a plurality of core via 23 penetrating through the glass substrate 21 in a thickness direction; and a core distribution layer 24 disposed on a surface of the glass substrate or a surface of the core via, and where an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, is disposed.

[0095] The glass substrate 21 has a first surface 213 and a second surface 214 facing each other, and the two surfaces are substantially parallel to each other and have a substantially uniform thickness throughout the glass substrate.

[0096] A core via 23 penetrating through the first surface and the second surface is disposed at the glass substrate 21.

[0097] Conventionally, a silicon substrate and an organic substrate were applied to the packaging substrate of the semiconductor apparatus, in a shape of being stacked. In case of a silicon substrate, when it is applied to a high-speed circuit, a parasitic element effect may occur due to its semiconductor property, and there is an advantage of relatively large power loss. Also, in case of an organic substrate, it requires a larger area to form a more complicated distribution pattern, but this does not correspond to the miniaturization trend of electronic devices. In order to form a complicated distribution pattern within a predetermined size, it is necessary to make patterns finer substantially, but

there has been a practical limit to the miniaturization of the patterns due to a material property of the polymer, etc., applied to an organic substrate.

[0098] In the embodiment, the glass substrate 21 is applied as a supporting body for the core layer 22 to solve these problems. Also, by applying a glass substrate and the core via 23 formed to penetrating through the glass substrate, it is possible to provide a packaging substrate 20 having a shortened electrical flow length, a smaller size, a faster response, and a lower loss property.

[0099] As the glass substrate 21, a glass substrate applied to semiconductor can be applied. For example, a borosilicate glass substrate, a non-alkali glass substrate, or the like may be applicable, but not limited thereto.

[0100] The glass substrate 21 may have a thickness of 1,000 μm or less, 100 to 1,000 μm , or 100 to 700 μm . More specifically, the glass substrate 21 may have a thickness of 100 to 500 μm . Although applying a thinner packaging substrate is advantageous in that electrical signal transmission can be made more efficient, but the packaging substrate also should serve as a supporting body of packaging, so it is preferable to apply the glass substrate 21 having the above thickness. Here, the thickness of the glass substrate may be the thickness of the glass substrate itself except for the thickness of an electrically conductive layer on the glass substrate.

[0101] The core via 23 may be formed by removing a predetermined region of the glass substrate 21. In detail, it may be formed by etching a glass plate physically and/or chemically.

[0102] In detail, the core via 23 may be formed by applying a method of forming a defect (flaw) on the surface of the glass substrate by means of a laser or the like and then chemical etching, laser etching, or the like, but not limited thereto.

[0103] The core via 23 includes a first opening part 233 in contact with the first surface; a second opening part 234 in contact with the second surface; and a minimum inner diameter part 235 having the smallest inner diameter in the entire core via connecting the first opening part and the second opening part.

[0104] A diameter CV1 of the first opening part and a diameter CV2 of the second opening part may substantially differ, or a diameter CV1 of the first opening part and a diameter CV2 of the second opening part may be substantially equal.

[0105] The minimum inner diameter part may be disposed in the first opening part or the second opening part. In this case, a core via may be a cylindrical-type or a (truncated) trigonal-pyramid-type. In this case, a diameter CV3 of the minimum inner diameter part corresponds to a diameter of the smaller one between the first opening part and the second opening part.

[0106] The minimum inner diameter part may be disposed between the first opening part and the second opening part. In this case, the core via may be a barrel-type core via. In this case, the diameter CV3 of the minimum inner diameter part may be smaller than a larger one between a diameter of the first opening part and a diameter of the second opening part.

[0107] The core distribution layer 24 includes a core distribution pattern 241, which is an electrically conductive layer electrically connecting the first surface and the second

surface of the glass substrate through a through-via, and a core insulating layer **223** surrounding the core distribution pattern.

[0108] The core layer **22** has an electrically conductive layer formed therein through a core via and thus serves as an electrical passage crossing the glass substrate **21**, connects an upper and a lower part of the glass substrate with a relatively short distance, and thereby may have properties of faster electrical signal transmission and lower loss.

[0109] The core distribution pattern **241** is a pattern that electrically connect the first surface **213** and the second surface **214** of the glass substrate through a core via **23**, and, specifically, it includes a first surface distribution pattern **241a**, which is an electrically conductive layer disposed on at least a part of the first surface **213**, a second surface distribution pattern **241c**, which is an electrically conductive layer disposed on at least a part of the second surface **214**, and a core via distribution pattern **241b**, which is an electrically conductive layer electrically connecting the first surface distribution pattern and the second surface distribution pattern to each other through the core via **23**. As the electrically conductive layers, for example, a copper plating layer may be applicable, but not limited thereto.

[0110] The core via **23** includes a first opening part **233** in contact with the first surface; a second opening part **234** in contact with the second surface; and a minimum inner diameter part **235** which is an area with the narrowest inner diameter in the entire core via connecting the first opening part and the second opening part.

[0111] The glass substrate **21** serves as an intermediate role or an intermediary role, connecting a semiconductor element unit **30** and a motherboard **10** to an upper and a lower part thereof, respectively, and the core via **23** serves as a transmitting passage for electrical signals thereof, thereby facilitating signal transmission.

[0112] A thickness of an electrically conductive layer measured at a larger one between the first surface opening part diameter and the second surface opening part diameter may be equal to or thicker than a thickness of an electrically conductive layer formed on a part with a minimum inner diameter in a core via.

[0113] The core distribution layer **24** is an electrically conductive layer formed on a glass substrate, and may satisfy that a cross-cut adhesion test value according to ASTM D3359 is 4 B or greater, and specifically may satisfy that the cross-cut adhesion test value is 5 B or greater. Also, an electrically conductive layer which is a core distribution layer **24**, may have an adhesive strength of 3 N/cm or more and a bonding strength of 4.5 N/cm or more with respect to the glass substrate **21**. When such a degree of bonding strength is satisfied, it has a sufficient bonding strength between a substrate and an electrically conductive layer, to be applied as a packaging substrate.

[0114] An upper layer **26** is disposed on the first surface **213**.

[0115] The upper layer **26** may include an upper distribution layer **25** and an upper surface connecting layer **27** disposed on the upper distribution layer **25**, and the uppermost surface of the upper layer **26** may be protected by a cover layer **60** having an opening part formed thereon, which is capable of being in direct contact with a connecting electrode of the semiconductor element unit.

[0116] The upper distribution layer **25** includes an upper insulating layer **253** disposed on the first surface; and an

upper distribution pattern **251** that has a predetermined pattern and is an electrically conductive layer at least a part of which is electrically connected to the core distribution layer **24**, and built in the upper insulating layer.

[0117] Anything applied as an insulating layer to a semiconductor element or a packaging substrate, is applicable to the upper insulating layer **253**, for example, an epoxy-based resin comprising a filler may be applied, but not limited thereto.

[0118] The insulating layer may be formed by a method of forming and hardening a coating layer, or by a method of laminating an insulating film which is being filmed in a state of non-hardened or semi-hardened to a core layer and hardening it. In this time, when a method of pressure sensitive lamination and the like is applied, the insulator is embedded even in the space inside a core via, and thus efficient process proceeding can be made. Also, even though plural-layered insulating layers are applied with being stacked, substantial distinction between the layers may be difficult, so that a plurality of insulating layer are collectively referred to as an upper insulating layer. Also, the core insulating layer **223** and the upper insulating layer **253** may be applied with the same insulating material, and in this case, the boundary therebetween may not be substantially distinguished.

[0119] The upper distribution pattern **251** refers to an electrically conductive layer disposed in the upper insulating layer **253** in a predetermined form. For example, it may be formed by a method of a build-up layer method. In detail, the upper distribution pattern **251** where electrically conductive layer is vertically or horizontally formed in a desired pattern, may be formed by repeating a process of: forming an insulating layer, removing an unnecessary part of the insulating layer and then forming an electrically conductive layer through a method of copper plating and the like, removing an unnecessary part of the electrically conductive layer and then forming an insulating layer on this electrically conductive layer again, and removing an unnecessary part again and then forming an electrically conductive layer through a method of plating and the like.

[0120] Since the upper distribution pattern **251** is disposed between the core layer **22** and the semiconductor element unit **30**, it is formed to at least partially includes a fine pattern so that the transmission of electrical signals with the semiconductor element unit **30** may proceed smoothly and a desired complicated pattern may be sufficiently accommodated. In this case, the fine pattern may have a width and an interval of about less than 4 μm , 3.5 μm or less, 3 μm or less, 2.5 μm or less, or 1 to 2.3 μm , respectively (Hereinafter, the description of the fine pattern is the same).

[0121] In order to form the upper distribution pattern **251** to include a fine pattern, at least two or more methods are applied in the present disclosure.

[0122] One of them, is to apply a glass substrate **21**, as a glass substrate **21** of a packaging substrate. The glass substrate **21** can have a considerably flat surface property with a surface roughness (Ra) of 10 angstroms or less, and thereby minimizing the influence of surface morphology of a supporting substrate on formation of the fine pattern.

[0123] The other one, is based on the property of the insulating layer. In case of the insulating layer, a filler component is often applied in addition to resin, and inorganic particles such as silica particles may be applicable as the filler. When the inorganic particles are applied to the

insulating layer as the filler, the size of the inorganic particles can affect whether to form the fine pattern, and therefore, the insulating layer in the present disclosure applies particle fillers with an average diameter of about 150 nm or less, and in detail, including particle fillers with an average diameter of 1 to 100 nm. Such a characteristic can minimize the influence of the insulating layer itself on the formation of an electrically conductive layer with a width of several micrometer-unit, while maintaining necessary properties for the insulating layer at a certain level or more, and can also help to form a fine pattern with good adhesion onto the surface, due to the fine surface morphology.

[0124] The upper surface connecting layer 27 includes an upper surface connecting pattern 272 disposed in the upper insulating layer 253, and at least a part of which is electrically connected to the upper distribution pattern 251, and an upper surface connecting electrode 271, electrically connecting the semiconductor element unit 30 and the upper surface connecting pattern 272. The upper surface connecting pattern 272 may be disposed on one surface of the upper insulating layer 253 or may be embedded with at least a part of which is being exposed on the upper insulating layer. For example, when the upper surface connecting pattern is disposed on one side of the upper insulating layer, the upper insulating layer may be formed by a method of plating and the like, and when the upper surface connecting pattern is embedded with at least a part of which is being exposed on the upper insulating layer, it may be the one which is formed by forming a copper plating layer and the like, and then a part of an insulating layer or electrically conductive layer is removed by a method of surface polishing, surface etching and the like.

[0125] The upper surface connecting pattern 272 may at least partially include a fine pattern like the above-described upper distribution pattern 251. The upper surface connecting pattern 272 including the fine pattern like this may enable a larger number of elements to be electrically connected to one another even in a narrow area, facilitate electrical signal connection between elements or with the external, and more integrated packaging is possible.

[0126] The upper surface connecting electrode 271 may be connected to the semiconductor element unit 30 directly through a terminal and the like or via an element connecting unit 51 such as a solder ball.

[0127] The packaging substrate 20 is also connected to the motherboard 10. The motherboard 10 may be directly connected to the second surface distribution pattern 241c, which is a core distribution layer disposed on at least a part of the second surface 214 of the core layer 22, through a motherboard terminal or may be electrically connected via a board connecting unit such as a solder ball. Also, the second surface distribution pattern 241c may be connected to the motherboard 10 via the lower layer 29 disposed under the core layer 22.

[0128] The lower layer 29 includes a lower distribution layer 291 and a lower surface connecting layer 292.

[0129] The lower distribution layer 291 includes i) a lower insulating layer 291b at least a part of which is in contact with the second surface 214; and ii) a lower distribution pattern 291a being embedded in the lower insulating layer and having a predetermined pattern, and at least a part of which is electrically connected to the core distribution layer.

[0130] The lower surface connecting layer 292 includes i) a lower surface connecting electrode 292a electrically con-

nected to the lower surface connecting pattern and may further include ii) a lower surface connecting pattern 292b at least a part of which is electrically connected to the lower distribution pattern, and at least a part of which is exposed to one surface of the lower insulating layer.

[0131] The lower surface connecting pattern 292b, which is a part connected to the motherboard 10, may be formed as a non-fine pattern wider than the fine pattern, unlike the upper surface connecting pattern 272, for more efficient transmitting of electrical signals.

[0132] Not applying a substantially additional different substrate other than the glass substrate 21 to the packaging substrate 20 disposed between the semiconductor element unit 30 and the motherboard 10, is one feature of the present disclosure.

[0133] Conventionally, an interposer and an organic substrate were applied with being stacked between connection of the element and the motherboard. It is considered that such a multi-stage form has been applied in at least two reasons. One reason is that there is a scale problem in directly bonding the fine pattern of the element to the motherboard, and the other reason is that problem of wiring damage may occur due to a difference in thermal expansion coefficient during the bonding process or during the driving process of the semiconductor apparatus. The embodiment has solved these problems by applying the glass substrate with a thermal expansion coefficient similar to that of the semiconductor element, and by forming a fine pattern with a fine scale enough to mount the elements on the first surface of the glass substrate and its upper layer.

[0134] In the embodiment of an electrically conductive layer composing the core via distribution pattern 214b, a thickness of the electrically conductive layer may be about 90% or more, about 93% to about 100%, or about 95% to about 100%, when a distance from an inner diameter surface of the core via 22 to a surface of the core via distribution pattern 214b is 100% as total. More specifically, in an electrically conductive layer composing the core via distribution pattern 214b, a thickness of the electrically conductive layer may be about 97% to 100%, or about 96% to about 100%, when a distance from an inner diameter surface of the core via 22 to a surface of the core via distribution pattern 214b is 100% as total.

[0135] Specifically, at a position of a minimum inner diameter, a thickness Tcv3 of the electrically conductive layer may be about 90% or more, about 93% to about 100%, about 95% to about 100%, or about 95.5% to about 99%, when a distance Dt3 from an inner diameter surface of the core via 22 to a surface of the core via distribution pattern 214b is 100% as total.

[0136] Specifically, at an opening part where a larger one between the first surface opening part diameter and the second surface opening part diameter is disposed, a thickness Tcv2 of the electrically conductive layer may be about 90% or more, about 93% to about 100%, about 95% to about 100%, or about 98% to about 99.9%, when a distance Dt2 (Assuming the second surface opening part diameter is a larger one) from an inner diameter surface of the core via 22 to a surface of the core via distribution pattern 214b is 100% as total.

[0137] When a thickness of the core via distribution pattern which is an electrically conductive layer, is formed substantially thickly inside the core via like this, more smooth electrical conductivity can be obtained, and a pro-

cess of connecting electrical signals to a motherboard from a highly integrated element can be made more efficient.

[0138] Also, one feature of the embodiment is that an electrically conductive layer formed on inner diameter surface of core via, is positioned close to a surface of a glass substrate (inner diameter surface of core via).

[0139] Specifically, the core via pattern may have an average distance of about 1 μm or less, about 0.9 μm , or about 0.01 μm to about 0.9 μm , between i) one surface of the core via pattern close to an inner diameter surface of the core via; and ii) the inner diameter surface of the core via.

[0140] An inorganic material or an organic-inorganic composite material which improves adhesion between the core via pattern and the glass surface or serves as a seed during a plating process, may be disposed in a space corresponding to the above distance. The organic-inorganic composite material may have a layered structure of being continuously observed in the cross section, and in this case an average thickness of the layered structure may correspond to the average distance. The organic-inorganic composite material may have a spotty structure of not being continuously observed in the cross section, and in which the shapes of particles are being located with an interval regularly or irregularly. When the organic-inorganic composite material has the spotty structure, the distance refers to an average value.

[0141] At the minimum inner diameter part, the core via pattern may have a distance F3 of about 1 μm or less, about 0.7 μm or less, or about 0.01 μm to about 0.7 μm , between i) one surface of the core via pattern close to an inner diameter surface of the core via; and ii) the inner diameter surface of the core via. An organic layer, an organic-inorganic composite layer, an adhesive layer, and the like with a thickness of about 1 μm or more, may not be substantially formed separately between an inner diameter surface and an electrically conductive layer, in the distance F3.

[0142] At the first opening part, the core via pattern may have a distance (not shown) of about 1 μm or less, about 0.9 μm or less, or about 0.01 μm to about 0.9 μm , between i) one surface of the core via pattern close to an inner diameter surface of the core via; and ii) the inner diameter surface of the core via. An adhesive layer and the like with a thickness of about 1 μm or more, may not be substantially formed separately between an inner diameter surface and an electrically conductive layer, in the above distance.

[0143] At the second opening part, the core via pattern may have a distance F2 of about 1 μm or less, about 0.9 μm or less, or about 0.01 μm to about 0.9 μm , between i) one surface of the core via pattern close to an inner diameter surface of the core via; and ii) the inner diameter surface of the core via. An adhesive layer and the like with a thickness of about 1 μm or more, may not be substantially formed separately between an inner diameter surface and an electrically conductive layer, in the distance F2.

[0144] The distance (not shown) at the first opening part, the distance F2 at the second opening part, and the distance F3 at the minimum inner diameter part respectively refer to a distance measured at an extension line of the first surface of the glass substrate, a distance measured at an extension line of the second surface of the glass substrate, and a distance measured at a virtual line substantially parallel to the first surface or the second surface of the glass at the minimum inner diameter part.

[0145] Within the core via, when a core via distribution pattern, which is an electrically conductive layer, is formed close to an inner diameter surface like this, a substantially thick electrically conductive layer can be manufactured under the same conditions, a more facilitated electrical conductivity as desired can be achieved, and a process of connecting electrical signals from a highly integrated element to a motherboard can be made more efficient.

[0146] The semiconductor apparatus 100 having a considerably thin packaging substrate 20 may make the overall thickness of the semiconductor apparatus thinner, and it is also possible to dispose a desired electrical connecting pattern even in a narrower area by applying the fine pattern. In detail, the packaging substrate may have a thickness of about 2000 μm or less, about 1500 μm or less, or about 900 μm . Also, the packaging substrate 20 may have a thickness of about 120 μm or more, or about 150 μm or more. Due to the above-described characteristics, the packaging substrate can stably connect the element and the motherboard electrically and structurally even with a relatively thin thickness, thereby contributing to miniaturization and thinning of the semiconductor apparatus.

[0147] The packaging substrate 20 may have a resistance value of about $2.6 \times 10^{-6} \Omega$ or more, about $3.6 \times 10^{-6} \Omega$ or more, or about $20.6 \times 10^{-6} \Omega$ or more, based on a cut one of the packaging substrate 20 into 100 $\mu\text{m} \times 100 \mu\text{m}$ as an upper surface size. The packaging substrate may have a resistance value of about $27.5 \times 10^{-6} \Omega$ or less, about $25.8 \times 10^{-6} \Omega$ or less, or about $24.1 \times 10^{-6} \Omega$ or less. For example, the resistance value is a measured resistance between an electrically conductive layer of an upper layer, and an electrically conductive layer of a lower layer, of the cut one of the packaging substrate into the constant size described above. And it is a resistance value measured by connecting the electrically conductive layer of the upper layer, and the electrically conductive layer of the lower layer to each other, by core via pattern. The resistance value may be measured by methods described at below experimental examples. Packaging substrate satisfying the above resistance value is possible to easily transmit electrical signals to external.

[0148] FIGS. 7 to 9 are flowcharts for illustrating a process of manufacturing a packaging substrate by cross sections thereof according to the embodiment. A method of manufacturing the packaging substrate according to another embodiment will be described below with reference to FIGS. 7 to 9.

[0149] The method of manufacturing the packaging substrate of the embodiment includes a preparation step of forming a defect at predetermined positions of a first surface and a second surface of a glass substrate; an etching step of preparing a glass substrate with a core via formed thereon by applying an etchant to the glass substrate where the defect is formed; a core layer forming step of plating the surface of the glass substrate with the core via formed thereon, to form a core distribution layer which is an electrically conductive layer, and thereby forming a core layer; and an upper layer forming step of forming an upper distribution layer, which is an electrically conductive layer surrounded by an insulating layer on one side of the core layer, and thereby manufacturing the packaging substrate described above.

[0150] The core layer forming step may include a pretreatment process of preparing a pretreated glass substrate by forming an organic-inorganic composite primer layer containing a nanoparticle with amine-group on a surface of

the glass substrate where the core via is formed; and a plating process of plating a metal layer on the glass substrate which is pretreated.

[0151] The core layer forming step may include a pre-treatment process of preparing a pretreated glass substrate by forming a metal-containing primer layer through sputtering on a surface of the glass substrate where the core via is formed; and a plating process of plating a metal layer on the glass substrate which is pretreated.

[0152] An insulating layer forming step may be further included between the core layer forming step and the upper layer forming step.

[0153] The insulating layer forming step may be a step of positioning an insulating film on the core layer and performing pressure sensitive lamination to form a core insulating layer.

[0154] The method of manufacturing the packaging substrate will be described in more detail.

[0155] 1) Preparation Step (Glass Defect Forming Process): A glass substrate **21a** having flat first surface and second surface was prepared, and a defect (groove) **21b** was formed at a predetermined position on the surface of the glass substrate to form a core via. As the glass substrate, a glass substrate applied to a substrate for electronic apparatus, etc., for example, non-alkalic glass substrate, etc. is applicable, but not limited thereto. As a commercial product, product manufactured by manufacturers such as CORNING, SCHOTT, AGC may be applied. For formation of the defect (groove), a method of mechanical etching, laser irradiation, and the like can be applied.

[0156] 2) Etching Step (Core Via Forming Step): On the glass substrate **21a** where the defect (groove) **21b** is formed, a core via **23** is formed through a physical or chemical etching process. During the etching process, the glass substrate forms vias in the parts with defect, and at the same time, the surface of the glass substrate **21a** may be simultaneously etched. A masking film may be applied to prevent the etching of the glass surface, but the defective glass substrate itself may be etched in consideration of the inconvenience, etc. of the process of applying and removing the masking film, and in this case, a thickness of the glass substrate having the core via may be slightly thinner than the thickness of the first glass substrate.

[0157] 3-1) Core Layer Forming Step: An electrically conductive layer **21d** is formed on the glass substrate. As for the electrically conductive layer, a metal layer containing copper metal may be applied representatively, but not limited thereto.

[0158] A surface of the glass (including a surface of a glass substrate and a surface of a core via) and a surface of the copper metal have different properties, so the adhesion strength is rather poor. In the embodiment, the adhesion strength between the glass surface and the metal is improved by two methods, a dry method and a wet method.

[0159] The dry method is a method applying sputtering, that is, a method of forming a seed layer **21c** inside the core via and on the glass surface through metal sputtering. For the formation of the seed layer, different kinds of metals such as titanium, chromium, and nickel may be sputtered with copper, etc., and in this case, it is considered that the adhesiveness of glass-metal is improved by surface morphology of glass, an anchor effect which is an interaction between metal particles, and the like.

[0160] The wet method is a method applying primer treatment, that is, a method of forming a primer layer **21c** by performing pre-treatment with a compound having a functional group such as amine. After pre-treatment by using a silane coupling agent depending on the degree of intended adhesion strength, primer treatment may be done with a compound or particles having an amine functional group. As mentioned above, a supporting body substrate of the embodiment needs to be of high performance enough to form a fine pattern, and it should be maintained after the primer treatment. Therefore, when such a primer includes a nanoparticle, it is desirable to apply a nanoparticle with an average diameter of 150 nm or less, for example, a nanoparticle is desirable to be applied to a particle with amine functional group. The primer layer may be formed by applying an adhesive strength improving agent manufactured in CZ series by MEC Inc, for example.

[0161] In the seed layer/primer layer **21c**, an electrically conductive layer may selectively form a metal layer in the state of removing a part where the formation of an electrically conductive layer is unnecessary, or not removing. Also, in the seed layer/primer layer **21c**, a part where the formation of an electrically conductive layer is necessary, or a part where it is unnecessary, may be selectively processed to be an activated state or an inactivated state for metal plating. The processing to be an activated state or an inactivated state may be performed, for example, by using light irradiation treatment such as laser light of a certain wavelength, etc., chemical treatment, and the like. A copper plating method, etc. applied to manufacturing a semiconductor element may be applied to form the metal layer, but not limited thereto.

[0162] During the metal plating, a thickness of an electrically conductive layer formed, may be controlled by regulating several variables such as the concentration of plating solution, plating time, and type of additive to be applied.

[0163] When a part of the core distribution layer is unnecessary, it may be removed, and an etched layer **21e** of a core distribution layer may be formed by performing metal plating to form an electrically conductive layer as a predetermined pattern, after the seed layer is partially removed or processed to be inactivated.

[0164] 3-2) Insulating Layer Forming Step: An insulating layer forming step in which an empty space of a core via is filled with an insulating layer after the core distribution layer, which is an electrically conductive layer, is formed, may be performed. In this case, the one manufactured in a film type may be applied to the applied insulating layer, and for example, a method such as pressure sensitive laminating the film-type insulating layer may be applied. When proceeding the pressure sensitive laminating like this, the insulating layer may be sufficiently subsided to the empty space inside the core via to form a core insulating layer without void formation.

[0165] 4) Upper Layer Forming Step: It is a step of forming an upper distribution layer including an upper insulating layer and an upper distribution pattern on a core layer. The upper distribution layer may be formed by a method of coating a resin composition forming an insulating layer **23a**, or laminating an insulating film. For simplicity, applying a method of laminating an insulating film is desirable. The laminating of the insulating film may be proceeded by a process of laminating and then hardening, and in this case, if a method of the pressure sensitive lamination is applied, the insulating resin may be sufficiently

subsidied even into a layer where an electrically conductive layer is not formed inside the core via. The upper insulating layer is also in direct contact with a glass substrate at least in part thereof, and thus the one with a sufficient adhesive force is applied. Specifically, it is desirable that the glass substrate and the upper insulating layer have characteristics that satisfy an adhesion strength test value of 4 B or more according to ASTM D3359.

[0166] The upper distribution pattern may be formed by repeating a process of forming the insulating layer **23a**, forming an electrically conductive layer **23c** to have a predetermined pattern, and forming an etched layer **23d** of the electrically conductive layer by etching the unnecessary part, and in the case of an electrically conductive layer formed to neighbor with having an insulating layer disposed therebetween, it may be formed by a method of performing a plating process after forming a blind via **23b** in the insulating layer. For formation of the blind via, a dry etching method such as laser etching and plasma etching, and a wet etching method using a masking layer and an etching solution may be applied.

[0167] 5) Upper Surface Connecting Layer and Cover Layer Forming Step: Upper surface connecting pattern and upper surface connecting electrode may be performed by a process similar to forming the upper distribution layer. Specifically, it may be formed by a method such as forming an etched layer **23f** of an insulating layer **23e** on the insulating layer **23e**, and then forming an electrically conductive layer **23g** again thereon, and then forming an etched layer **23h** of the electrically conductive layer, but a method of selectively forming only an electrically conductive layer without applying a method of etching, may be also applied. A cover layer may be formed to have an opening part (not shown) at a position corresponding to the upper surface connecting electrode such that the upper surface connecting electrode to be exposed and directly connected to an element connecting unit, a terminal of an element, or the like.

[0168] 6) Lower Surface Connecting Layer and Cover Layer Forming Step: A lower distribution layer and/or a lower surface connecting layer, and optionally a cover layer (not shown) may be formed in a manner similar to the the upper surface connecting layer and the cover layer forming step, described above.

[0169] Hereinafter, the embodiment will be described in more detail through specific examples. The following examples are only examples to help the understanding of the embodiment, and the scope of the embodiment is not limited thereto.

Example 1—Manufacturing of a Packaging Substrate

[0170] 1) Preparation Step (Glass Defect Forming Process): A glass substrate **21a** with a flat first surface and a flat second surface was prepared, and defects (grooves, **21b**) were formed on the glass surface at a predetermined position for forming a core via. In this case, the number of defects were allowed to form in the number of 225 per 1 cm². As the glass, borosilicate glass (from CORNING) was applied. A method of mechanical etching and laser irradiation were applied to the formation of the defect (groove).

[0171] 2) Etching Step (Core Via Forming Step): The core via **23** was formed on the glass substrate **21a** where the defects (grooves, **21b**) were formed, through a physical or chemical etching process. In this case, the core via were

formed to have a first opening part in contact with the first surface; a second opening part in contact with the second surface; and a minimum inner diameter part, which is the area whose diameter is the narrowest in the entire core via connecting the first opening part and the second opening part.

[0172] 3-1) Core Layer Forming Step: An electrically conductive layer **21d** was formed on a glass substrate. As the electrically conductive layer, a metal layer containing copper metal was applied. By two methods of a dry method and a wet method, an adhesive strength between a surface of the glass substrate and a metal layer was improved. The dry method is a method applying sputtering, that is, a method of forming a seed layer **21c** inside the core via and on the glass surface through metal sputtering. For the formation of the seed layer, one or more different kinds of metals from titanium, chromium, and nickel were sputtered with copper, etc. The wet method is a method applying primer treatment, that is, a method of forming a primer layer **21c** by performing pre-treatment with a compound having a functional group such as amine. After pre-treatment by using a silane coupling agent, primer treatment was done with a compound or particles having an amine functional group. A nanoparticle with an average diameter of 150 nm or less was applied to this primer, and a nanoparticle was applied to a particle with amine functional group. The primer layer was formed by applying an adhesive strength improving agent manufactured in CZ series by MEC Inc.

[0173] In the seed layer/primer layer **21c**, a part where the formation of an electrically conductive layer is necessary, or a part where it is unnecessary, was selectively processed to be an activated state or an inactivated state for metal plating. Light irradiation treatment such as laser light of a certain wavelength, etc., chemical treatment, and the like were applied to the processing to be an activated state or an inactivated state. A copper plating method applied to manufacturing a semiconductor element was applied to form the metal layer.

[0174] An etched layer **21e** of a core distribution layer was formed by performing metal plating to form an electrically conductive layer as a predetermined pattern, after the seed layer is partially removed or processed to be inactivated. At a position of the minimum inner diameter of the core via, the electrically conductive layer was allowed to have a thickness of 97%, when a distance from an inner diameter surface of the core via to a surface of the electrically conductive layer is total 100%. Also, at an opening part where has a larger diameter between the first opening part and the second opening part, the electrically conductive layer was allowed to have a thickness of 97%, when a distance from an inner diameter surface of the core via to a surface of the electrically conductive layer is total 100%. Furthermore, an average distance between one surface of the electrically conductive layer near to an inner diameter surface of the core via and the inner diameter surface of the core via, was allowed to be 0.5 μ m.

[0175] 3-2) Insulation Layer Forming step: After forming the core distribution layer, which is an electrically conductive layer, an insulating layer forming step of filling the empty space with an insulating layer was performed. At this time, the one manufactured in the form of a film was applied to the applied insulating layer, and a method of pressure sensitive lamination of the insulating layer in the form of a film was applied.

[0176] 4) Upper Layer Forming Step: A step of forming an upper distribution layer including an upper insulating layer and an upper distribution pattern on the core layer was performed. A method of laminating an insulating film as the upper insulating layer was performed, and a process of lamination and hardening of the insulating was performed. The upper insulating layer is also at least partially in direct contact with the glass substrate, and thus the one with a sufficient adhesive force was applied to. Specifically, the one having properties that satisfy an adhesion test value of 4 B or more according to ASTM D3359, were applied to the glass substrate and the upper insulating layer

[0177] The upper distribution pattern was formed by repeating the process of forming the insulating layer 23a, forming an electrically conductive layer 23c in a predetermined pattern, and etching unnecessary parts to form an etching layer 23d of an electrically conductive layer. In the case of an electrically conductive layer formed adjacent to each other with an insulating layer disposed therebetween, it was formed by a method of forming a blind via 23b in the insulating layer and then performing a plating process. For the formation of the blind via, a dry etching method such as laser etching and plasma etching, and a wet etching method using a masking layer and an etchant were applied to manufacture a packaging substrate.

[0178] 5) Upper Surface Connecting Layer and Cover Layer Forming Step: A method of forming an etched layer 23f of an insulating layer 23e on the insulating layer, and then forming an electrically conductive layer 23g again thereon, and then forming an etched layer 23h of the electrically conductive layer, was proceeded. A cover layer was formed to have an opening part (not shown) at a position corresponding to the upper surface connecting electrode such that the upper surface connecting electrode to be exposed and directly connected to an element connecting unit, a terminal of an element, or the like.

[0179] 6) Lower Surface Connecting Layer and Cover Layer Forming Step: A lower distribution layer and/or a lower surface connecting layer, and optionally a cover layer (not shown) was formed in a manner similar to the upper surface connecting layer and the cover layer forming step, described above.

[0180] A packaging substrate 20 manufactured by the above methods comprises:

[0181] a core layer including a glass substrate 21 with a first surface and a second surface facing each other, a plurality of core via 23 penetrating through the glass substrate in a thickness direction, and a core distribution layer 24 disposed on a surface of the glass substrate or the core via, and where an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, is disposed; and

[0182] an upper layer 26 disposed on the first surface and includes an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

[0183] wherein the upper layer includes an upper distribution layer 25 and an upper surface connecting layer 27 disposed on the upper distribution layer,

[0184] the upper distribution layer includes an upper insulating layer 253 disposed on the first surface; and an upper distribution pattern 251 that has a predetermined pattern and is an electrically conductive layer at least a part of which is

electrically connected to the core distribution layer 24, and built in the upper insulating layer,

[0185] the core via includes a first opening part 233 in contact with the first surface; a second opening part 234 in contact with the second surface; and a minimum inner diameter part 235 having the smallest inner diameter in the entire core via connecting the first opening part and the second opening part,

[0186] wherein a thickness Tcv3 of an electrically conductive layer of the core distribution layer is 97%, when a distance Dt3 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total,

[0187] wherein at an opening part where a larger diameter between the first surface opening part and the second surface opening part is disposed, a thickness Tcv2 of an electrically conductive layer of the core distribution layer is 97%, when a distance Dt2 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total, and

[0188] wherein an average distance F3, between one surface of an electrically conductive layer of the core distribution layer close to an inner diameter surface of the core via and the inner diameter surface of the core, via is 0.5 μm .

Example 2—Manufacturing of a Packaging Substrate

[0189] In a packaging substrate of Example 1, a packaging substrate was manufactured by the same process with Example 1, except for changing a thickness Tcv3 of an electrically conductive layer of the core distribution layer to 95%, when a distance Dt3 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total,

[0190] a thickness Tcv2 of an electrically conductive layer of the core distribution layer to 95%, at an opening part where a larger diameter between the first surface opening part and the second surface opening part is disposed, when a distance Dt2 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via distribution layer is 100% as total, and

[0191] an average distance F3 to 0.75 μm , between one surface of an electrically conductive layer of the core distribution layer close to an inner diameter surface of the core via, and the inner diameter surface of the core via.

Example 3—Manufacturing of a Packaging Substrate

[0192] In a packaging substrate of Example 1, a packaging substrate was manufactured by the same process with Example 1, except for changing a thickness Tcv3 of an electrically conductive layer of the core distribution layer to 93%, when a distance Dt3 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total,

[0193] a thickness Tcv2 of an electrically conductive layer of the core distribution layer to 93%, at an opening part where a larger diameter between the first surface opening part and the second surface opening part is disposed, when a distance Dt2 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via distribution layer is 100% as total, and

[0194] an average distance F3 to 1 μm , between one surface of an electrically conductive layer of the core distribution layer close to an inner diameter surface of the core via, and the inner diameter surface of the core via.

Example 4—Manufacturing of a Packaging Substrate

[0195] In a packaging substrate of Example 1, a packaging substrate was manufactured by the same process with Example 1, except for changing a thickness Tcv3 of an electrically conductive layer of the core distribution layer to 92%, when a distance Dt3 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total,

[0196] a thickness Tcv2 of an electrically conductive layer of the core distribution layer to 92%, at an opening part where a larger diameter between the first surface opening part and the second surface opening part is disposed, when a distance Dt2 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via distribution layer is 100% as total, and

[0197] an average distance F3 to 1.2 μm , between one surface of an electrically conductive layer of the core distribution layer close to an inner diameter surface of the core via, and the inner diameter surface of the core via.

Example 5—Manufacturing of a Packaging Substrate

[0198] A packaging substrate was manufactured by the same process with Example 1, except for changing a thickness Tcv3 of an electrically conductive layer of the core distribution layer to 87%, when a distance Dt3 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total,

[0199] a thickness Tcv2 of an electrically conductive layer of the core distribution layer to 87%, at an opening part where a larger diameter between the first surface opening part and the second surface opening part is disposed, when a distance Dt2 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via distribution layer is 100% as total, and

[0200] an average distance F3 to 2 μm , between one surface of an electrically conductive layer of the core distribution layer close to an inner diameter surface of the core via, and the inner diameter surface of the core via, in the Example 1.

Experimental Example—Measurement of Electrical Properties

[0201] A packaging substrate of the Examples 1 to 5 was cut into the size of 100 μm ×100 μm based on an upper surface, and a resistance value among electrical properties were measured by resistivity meter. And the results are shown in Table 1.

TABLE 1

	Example 1	Example 2	Example 3	Example 4	Example 5
Tcv3/Dt3	0.97	0.95	0.93	0.92	0.87
Tcv2/Dt2	0.97	0.95	0.93	0.92	0.87
F3	0.5 μm	0.75 μm	1 μm	1.2 μm	2 μm

TABLE 1-continued

	Example 1	Example 2	Example 3	Example 4	Example 5
Bulk resistance value (Ω)	25.8×10^{-6}	24.5×10^{-6}	24.1×10^{-6}	23.7×10^{-6}	22.4×10^{-6}

[0202] Referring to Table 1, a packaging substrate of Examples 1 to 3 in which a thickness Tcv3 of an electrically conductive layer of the core distribution layer is 90% or more, when a distance Dt3 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core distribution layer is 100% as total, a thickness Tcv2 of an electrically conductive layer of the core distribution layer is 90% or more, at an opening part where a larger diameter between the first surface opening part and the second surface opening part is disposed, when a distance Dt2 from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via distribution layer is 100% as total, and an average distance F3 is 1 μm or less, between one surface of an electrically conductive layer of the core distribution layer close to an inner diameter surface of the core via, and the inner diameter surface of the core via, showed resistance values of $24.1 \times 10^{-6}\Omega$ to $25.8 \times 10^{-6}\Omega$, which are good resistance values. It is considered that a packaging substrate with these characteristics is possible to sufficiently easily transmit electrical signals to the element respectively disposed at above or below it.

[0203] A packaging substrate of the embodiment does not form a parasitic element which a glass substrate has, and with excellent characteristics such as being capable of serving as a supporting substrate which is thin and having sufficient strength, it forms an electrically conductive layer with an appropriate ratio of the thickness of the glass substrate, and thereby utilizes that excellent properties such as inducing efficient transmitting of signals.

[0204] A glass substrate is evaluated to have poor bonding properties with an electrically conductive layer such as a copper layer, and in order to form an electrically conductive layer with a sufficient thickness by a method of plating, etc., a seed layer, a primer layer or the like is needed to be formed between a glass surface and an electrically conductive layer. However, when forming such seed layer or primer layer too thick, it may be difficult to form an electrically conductive layer enough within the determined diameter of core via, and it may adversely affect the upper and lower electrical signal transmission rate of the packaging substrate.

[0205] Considering these characteristics, and for efficient electrical signal transmission, it is considered that applying a thickness of a seed layer or a primer layer as thin as possible while satisfying specific ratio, the Tcv3/Dt3 ratio and the Tcv2/Dt2 ratio of 0.90 or more, and F3 of 1.0 μm or less, is desirable.

[0206] Although the desirable examples of the embodiment have been described above, the scope of the embodiment is not limited thereto, and various modifications and alterations made by those skilled in the art using the basic concept of the embodiment defined in the following claims also fall within the scope of the embodiment.

DESCRIPTION OF FIG. NUMBERS	
100:	Semiconductor apparatus
10:	Motherboard
30:	Semiconductor element unit
32:	First semiconductor element
34:	Second semiconductor element
36:	Third semiconductor element
20:	Packaging substrate
22:	Core layer
223:	Core insulating layer
21, 21a:	Glass substrate
213:	First surface
214:	Second surface
23:	Core via
233:	First opening part
234:	Second opening part
235:	Minimum inner diameter part
24:	Core distribution layer
241:	Core distribution pattern
241a:	First surface distribution pattern
241b:	Core via distribution pattern
241c:	Second surface distribution pattern
26:	Upper layer
25:	Upper distribution layer
251:	Upper distribution pattern
252:	Blind via
253:	Upper insulating layer
27:	Upper surface connecting layer
271:	Upper surface connecting electrode
272:	Upper surface connecting pattern
29:	Lower layer
291:	Lower distribution layer
291a:	Lower distribution pattern
291b:	Lower insulating layer
292:	Lower surface connecting pattern
292a:	Lower surface connecting electrode
292b:	Lower surface connecting pattern
50:	Connecting part
51:	Element connecting part
52:	Board connecting part
60:	Cover layer
21b:	Glass defect
21c:	Seed layer, Primer layer
21d:	Core distribution layer
21e:	Etched layer of Core distribution layer
23a:	Insulating layer
23b:	Etched layer of Insulating layer
23c:	Electrically conductive layer
23d:	Etched layer of Electrically conductive layer
23e:	Insulating layer
23f:	Etched layer of Insulating layer
23g:	Electrically conductive layer
23h:	Etched layer of Electrically conductive layer

What is claimed is:

1. A semiconductor apparatus comprising:

a semiconductor element unit where one or more semiconductor elements are disposed; a packaging substrate electrically connected to the semiconductor element; and a motherboard electrically connected to the packaging substrate, transmitting electrical signals of the semiconductor element and external, and connecting each other;

wherein a packaging substrate comprises a core layer, and an upper layer disposed on the core layer,

wherein the core layer comprises a glass substrate and a core via,

the glass substrate is with a first surface and a second surface facing each other,

the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

the core layer comprises a core distribution layer disposed on a surface of the glass substrate or the core via,

the core distribution layer comprises an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, and

the upper layer is disposed on the first surface and comprises an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

wherein the core via comprises a first surface opening part diameter, a second surface opening part diameter, and a minimum inner diameter,

wherein the first surface opening part diameter is a diameter at an opening part in contact with the first surface,

the second surface opening part diameter is a diameter at an opening part in contact with the second surface, and the minimum inner diameter is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

wherein at a position of the minimum inner diameter, a thickness of the electrically conductive layer is 90% or more, when a distance from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via is 100% as total.

2. The semiconductor apparatus of claim 1,

wherein at an opening part where a larger one between the first surface opening part diameter and the second surface opening part diameter is disposed, a thickness of the electrically conductive layer is 90% or more, when a distance from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via distribution pattern is 100% as total.

3. The semiconductor apparatus of claim 1, further comprising:

an upper insulating layer and an upper distribution pattern,

wherein the upper insulating layer is an insulating layer disposed on the first surface,

the upper distribution pattern is an electrically conductive layer at least a part of which is electrically connected to the core distribution layer,

the upper distribution pattern is built in the upper insulating layer,

the upper distribution pattern is at least partially comprising a fine pattern, and

the fine pattern has a width and an interval of less than 4 μm , respectively.

4. The semiconductor apparatus of claim 1,

wherein the packaging substrate has a resistance value of about $27.5 \times 10^{-6} \Omega$ or less, based on a cut one of the packaging substrate into 100 $\mu\text{m} \times 100 \mu\text{m}$ as an upper surface size.

5. A packaging substrate comprising:

a core layer, and an upper layer disposed on the core layer, wherein the core layer comprises a glass substrate and a core via,

the glass substrate is with a first surface and a second surface facing each other,

the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

the core layer comprises a core distribution layer disposed on a surface of the glass substrate or the core via,

the core distribution layer comprises an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, and

the upper layer is disposed on the first surface and comprises an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

wherein the core via comprises a first surface opening part diameter, a second surface opening part diameter, and a minimum inner diameter,

wherein the first surface opening part diameter is a diameter at an opening part in contact with the first surface,

the second surface opening part diameter is a diameter at an opening part in contact with the second surface, and the minimum inner diameter is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

wherein at a position of the minimum inner diameter, a thickness of the electrically conductive layer is 90% or more, when a distance from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via is 100% as total.

6. A semiconductor apparatus comprising:

a semiconductor element unit where one or more semiconductor elements are disposed; a packaging substrate electrically connected to the semiconductor element; and a motherboard electrically connected to the packaging substrate, transmitting electrical signals of the semiconductor element and external, and connecting each other;

wherein a packaging substrate comprises a core layer, and an upper layer disposed on the core layer,

wherein the core layer comprises a glass substrate and a core via,

the glass substrate is with a first surface and a second surface facing each other,

the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

the core layer comprises a core distribution layer disposed on a surface of the glass substrate or the core via,

the core distribution layer comprises an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, and

the upper layer is disposed on the first surface and comprises an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

wherein the core distribution layer comprises a first surface core pattern, a second surface core pattern, and a core via pattern,

wherein the first surface core pattern is an electrically conductive layer disposed on at least a part of the first surface,

the second surface core pattern is an electrically conductive layer disposed on at least a part of the second surface, and

the core via pattern is an electrically conductive layer electrically connecting the first surface core pattern and the second surface core pattern through the core via, and

wherein the core via pattern has an average distance of 1 μm or less, between one surface of the core via pattern close to an inner diameter surface of the core via; and the inner diameter surface of the core via.

7. The semiconductor apparatus of claim 6,

wherein the core via has a first surface opening part diameter, a second surface opening part diameter and a minimum inner diameter,

wherein the first surface opening part diameter is a diameter at an opening part in contact with the first surface,

the second surface opening part diameter is a diameter at an opening part in contact with the second surface, and the minimum inner diameter is a diameter at the narrowest area disposed between the first surface opening part and the second surface opening part, and

wherein at a position of the minimum inner diameter, a thickness of the electrically conductive layer is 90% or more, when a distance from an inner diameter surface of the core via to a surface of an electrically conductive layer of the core via is 100% as total.

8. The semiconductor apparatus of claim 7,

wherein the packaging substrate has a resistance value of about $27.5 \times 10^{-6} \Omega$ or less, based on a cut one of the packaging substrate into $100 \mu\text{m} \times 100 \mu\text{m}$ as an upper surface size.

9. The semiconductor apparatus of claim 6, further comprising:

an upper insulating layer and an upper distribution pattern,

wherein the upper insulating layer is an insulating layer disposed on the first surface,

the upper distribution pattern is an electrically conductive layer at least a part of which is electrically connected to the core distribution layer,

the upper distribution pattern is built in the upper insulating layer,

the upper distribution pattern is at least partially including a fine pattern, and

the fine pattern has a width and an interval of less than 4 μm , respectively.

10. A packaging substrate comprising:

a core layer, and an upper layer disposed on the core layer, wherein the core layer comprises a glass substrate and a core via,

the glass substrate is with a first surface and a second surface facing each other,

the core via penetrating through the glass substrate in a thickness direction is disposed in a plural number,

the core layer comprises a core distribution layer disposed on a surface of the glass substrate or the core via,

the core distribution layer comprises an electrically conductive layer at least a part of which electrically connect an electrically conductive layer of the first surface and an electrically conductive layer of the second surface through the core via, and

the upper layer is disposed on the first surface and comprises an electrically conductive layer which electrically connect the core distribution layer and semiconductor element unit of external;

wherein the core distribution layer comprises a first surface core pattern, a second surface core pattern, and a core via pattern,

wherein the first surface core pattern is an electrically conductive layer disposed on at least a part of the first surface,

the second surface core pattern is an electrically conductive layer disposed on at least a part of the second surface, and

the core via pattern is an electrically conductive layer electrically connecting the first surface core pattern and the second surface core pattern through the core via, and

wherein the core via pattern has an average distance of 1 μm or less, between one surface of the core via pattern close to an inner diameter surface of the core via; and the inner diameter surface of the core via.

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